generating set from the main or auxiliary compressed air system through the nonreturn valve fitted in the emergency generator space.

Steering and stabilization

Where steering and/or stabilization of a craft is essentially dependent on one device as with a single rudder or pylon, which is itself dependent on the continuous availability of electric power, it should be served by at least two independent circuits one of which should be fed either from the emergency source of electric power or from an independent power source located in such a position as to be unaffected by fire or flooding affecting the main source of power. Failure of either supply should not cause any risk to the craft or passengers during switching to the alternative supply and

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SOURCE: MSC.36(63).pdf

as to be unaffected by fire or flooding affecting the main source of power. Failure of either supply should not cause any risk to the craft or passengers during switching to the alternative supply and such switching arrangements should meet the requirements in 5.2.5. These circuits should be provided with short circuit protection and an overload alarm.

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Protection against excess current may be provided, in which case it should be for not less than twice the full load current of the motor or circuit so protected, and should be arranged to accept the appropriate starting current with a reasonable margin. Where three-phase supply is used, an alarm should be provided in a readily observed position in the craft's operating compartment that will indicate failure of anyone of the phases.

12.5.3 Where such systems are not essentially dependent on the continuous availability of electric power but at least one alternative system, not dependent on the electric supply is installed, then the

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indicate failure of anyone of the phases.

12.5.3 Where such systems are not essentially dependent on the continuous availability of electric power but at least one alternative system, not dependent on the electric supply is installed, then the electrically powered or controlled system may be fed by a single circuit protected in accordance with 12.5.2.

12.5.4 The requirements of chapters 5 and 16 for power supply of the directional control system and stabilization system of the craft should be met.

12.6 Precautions against shock, fire and other hazards of electrical origin

12.6.1.1 Exposed metal parts of electrical machines or equipment which are not intended to be live but which are liable under fault conditions to become live should be earthed unless the machines or equipment are:

.1 supplied at a voltage not exceeding 55 V direct current or 55 V root mean square between conductors; auto-transformers should not be used for the purpose of achieving this voltage; or

.2 supplied at a voltage not exceeding 250 V by safety isolating transformers supplying only one consuming device; or

.3 constructed in accordance with the principle of double insulation.

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or
.2 supplied at a voltage not exceeding 250 V by safety isolating transformers supplying only one consuming device; or
.3 constructed in accordance with the principle of double insulation.

12.6.1.2 The Administration may require additional precautions for portable electrical equipment for use in confined or exceptionally damp spaces where particular risks due to conductivity may exist.

12.6.1.3 All electrical apparatus should be so constructed and so installed as not to' cause injury when handled or touched in the normal manner.

12.6.2 Main and emergency switchboards should be so arranged as to give easy access, as may be needed, to apparatus and equipment, without danger to personnel. The sides and the rear and, where necessary, the front of switchboards should be suitably guarded. Exposed live parts having voltages to earth exceeding a voltage to be specified by the Administration should not be installed on the front of such switchboards. Where necessary, nonconducting mats or gratings should be provided at the front and rear of the switchboard.

12.6.3 When a distribution system, whether primary or secondary, for power,

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switchboards. Where necessary, nonconducting mats or gratings should be provided at the front and rear of the switchboard.

12.6.3 When a distribution system, whether primary or secondary, for power, heating or lighting, with no connection to earth is used, a device capable of continuously monitoring the insulation level to earth and

12.6.4

of giving an audible or visual indication of abnormally low insulation values should be provided. For limited secondary distribution systems the Administration may accept a device for manual checking of the insulation level.

Cables and wiring

12.6.4.1 Except as permitted by the Administration in exceptional circumstances, all metal sheaths and armour of cables should be electrically continuous and should be earthed.

12.6.4.2 All electric cables and wiring external to equipment should be at least of a flame-retardant type and should be so installed as not to impair their original flame-retarding properties. Where necessary

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12.6.4.2 All electric cables and wiring external to equipment should be at least of a flame-retardant type and should be so installed as not to impair their original flame-retarding properties. Where necessary for particular applications, the Administration may permit the use of special types of cables such as radio frequency cables, which do not comply with the foregoing.

12.6.4.3 Cables and wiring serving essential or emergency power, lighting, internal communications or signals should so far as practicable be routed clear of machinery spaces and their casings and other high fire risk areas. Where practicable all such cables should be run in such a manner as to preclude their being rendered unserviceable by heating of the bulkheads that may be caused by a fire in an adjacent space.

12.6.4.4 Where cables which are installed in hazardous areas introduce the risk of fire or explosion in the event of an electrical fault in such areas, special precautions against such risks should be taken to the satisfaction of the Administration.

12.6.4.5 Cables and wiring should be installed and supported in such manner as to avoid chafing or other damage.

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such areas, special precautions against such risks should be taken to the satisfaction of the Administration.

12.6.4.5 Cables and wiring should be installed and supported in such manner as to avoid chafing or other damage.

12.6.4.6 Terminations and joints in all conductors should be so made as to retain the original electrical, mechanical, flame-retarding and, where necessary, fire-resisting properties of the cable.

12.6.5.1 Each separate circuit should be protected against short circuit and against ·overload, except as permitted in 12.5, or where the Administration may exceptionally otherwise permit.

12.6.5.2 The rating or appropriate setting of the overload protective device for each ·circuit should be permanently indicated at the location of the protective device.

12.6.5.3 When the protective device is a fuse it should be placed on the load side of the disconnect switch serving the protected circuit.

12.6.6 Lighting fittings should be so arranged as to prevent temperature

rises which could damage the cables and wiring, and to prevent surrounding material from becoming excessively hot.
12.6.7 All lighting and power circuits terminating in a bunker or cargo

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rises which could damage the cables and wiring, and to prevent surrounding material from becoming excessively hot.
12.6.7 All lighting and power circuits terminating in a bunker or cargo space should be provided with a multiple-pole switch outside the space for disconnecting such circuits.
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12.6.8.1 Accumulator batteries should be suitably housed, and compartments used primarily for their accommodation should be properly constructed and efficiently ventilated.
12.6.8.2 Electrical or other equipment which may constitute a source of